Nonlinear Dynamics of the Logistic Map and an Educational Interval-Mapping Text Demonstration

Course Project

Abstract

We study the logistic map $x_{n+1} = rx_n(1 - x_n)$ through a time series, bifurcation diagram, Lyapunov sweep, sensitivity experiment, trajectory-separation plot, and invariant-density estimate. At $r = 4$, the computed Lyapunov exponent is 0.693020, close to ln 2 = 0.693147; the mean of 500,000 retained samples is 0.4992599, close to 0.5. Four messages were recovered in a small interval-mapping demonstration. This demonstration is educational only: it is not a faithful reproduction of Baptista's 1998 cryptosystem and is not evidence of security.

Keywords: logistic map; bifurcation; Lyapunov exponent; invariant density; chaotic cryptography

1. Introduction and model

The logistic map is a compact nonlinear model in which the control parameter r changes long-term behaviour from stable fixed points to periodic and chaotic regimes. The single-orbit experiments use $r = 4.0$ and $x_0 = 0.213$. Baptista's original iteration-count idea provides historical context [1], but the submitted text code is only a classroom-scale demonstration.

2. Methods

The bifurcation scan uses 3,000 values in [2.8, 4.0], discards 1,000 iterations, and records 200 more per value. The Lyapunov sweep uses 3,000 values in [2.5, 4.0], 1,000 transient iterates, and 3,000 accumulation iterates. For the invariant density, 1,000 transient iterates are removed and 500,000 samples at $r = 4$ are retained. The reference density is rho(x) = [pi sqrt(x(1-x))]$^{-1}$.

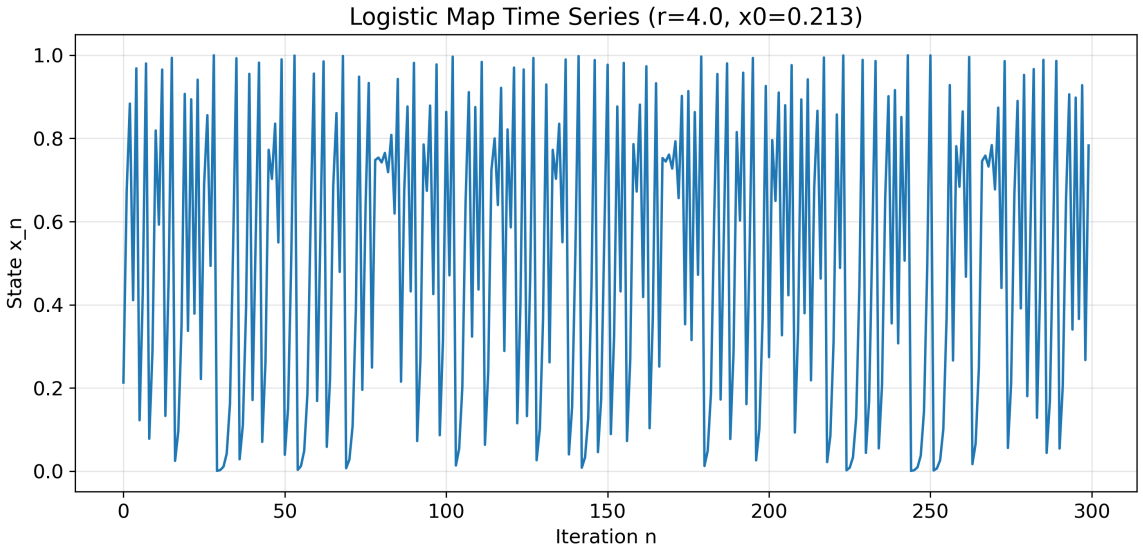


Figure 1. Logistic-map time series for $r = 4$ and $x_0 = 0.213$.

3. Parameter dependence and instability

The bifurcation diagram records long-term states while the Lyapunov curve estimates local exponential expansion or contraction. At $r = 4$, the numerical result is 0.693020 versus ln 2 = 0.693147. The nearby initial values 0.213 and 0.213 + 10^{-10} visibly separate under iteration.

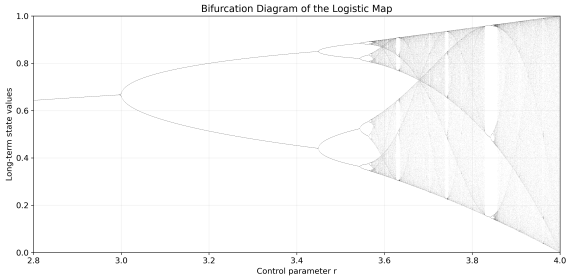


Figure 2. Bifurcation diagram, r in [2.8, 4.0].

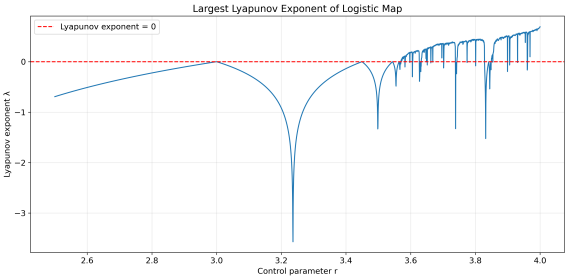


Figure 3. Numerical largest Lyapunov exponent.

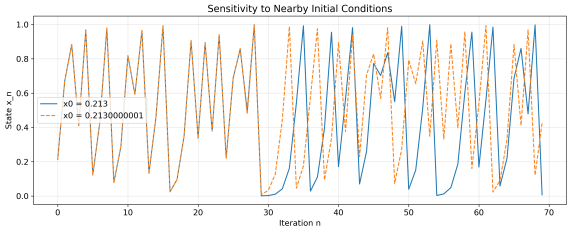


Figure 4. Sensitivity to a 10^{-10} perturbation.

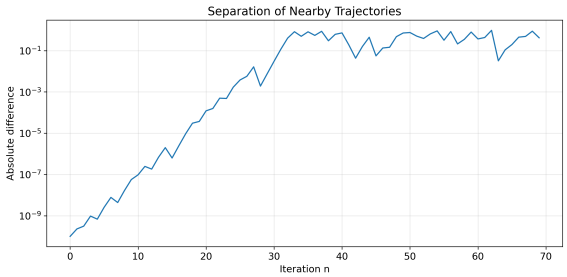


Figure 5. Absolute trajectory separation (log scale).

4. Invariant density and text demonstration

The empirical histogram has the expected U-shaped form. Its retained-sample mean, 0.4992599, is close to the theoretical mean 0.5. This is a numerical consistency check, not a cryptographic property.

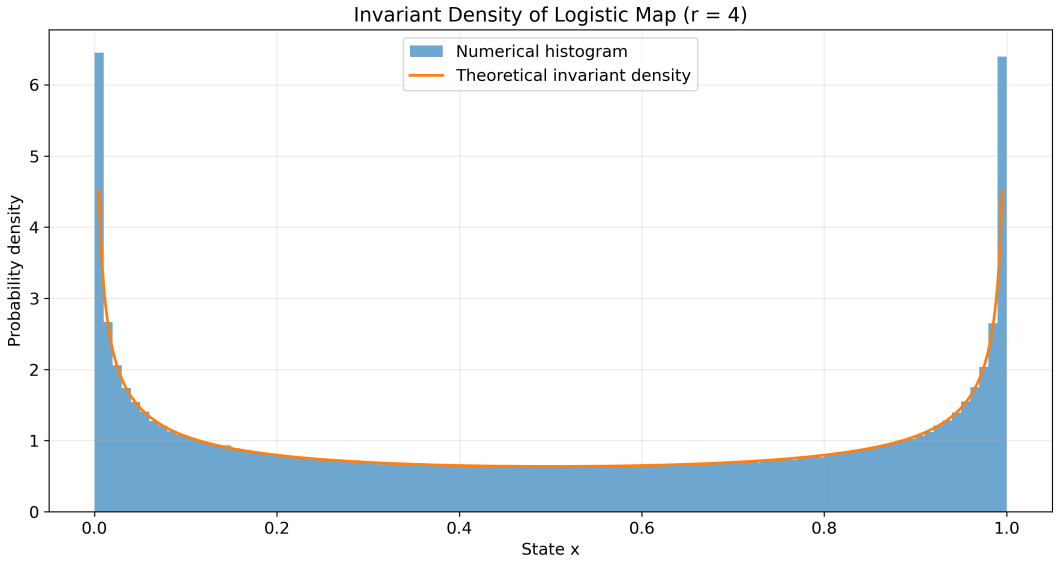


Figure 6. Numerical histogram and theoretical invariant density at $r = 4$.

The submitted code divides [0, 1] into 26 equal-width intervals for A-Z. It records a count until each target interval is reached, then replays the counts from the same initial state. CHAOS, LOGISTICMAP, NONLINEARDYNAMICS, and ENCRYPTION were recovered exactly (43 characters total).

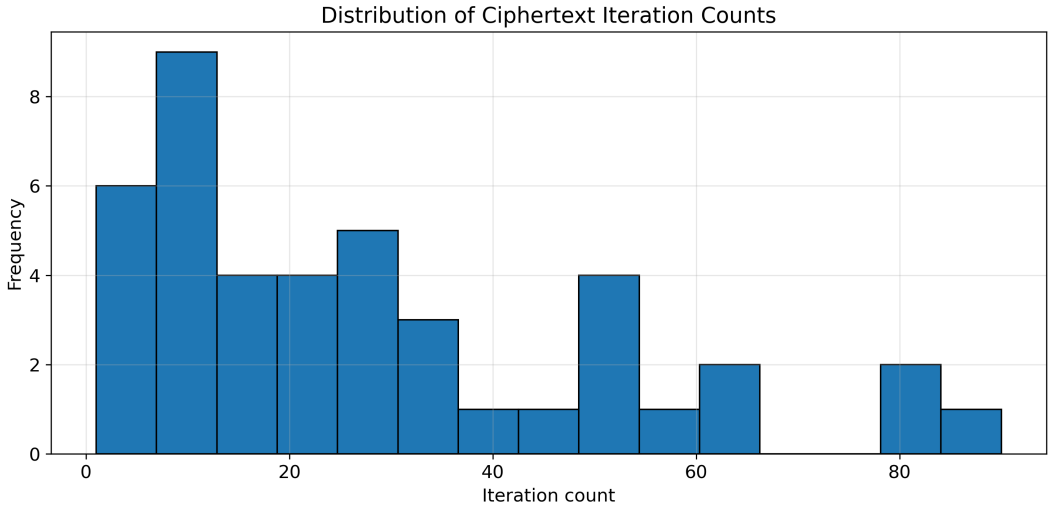


Figure 7. Iteration-count distribution for the four successful tests.

5. Literature context, limitations, and conclusion

Baptista proposed an iteration-count scheme based on logistic-map ergodicity [1]. The submitted implementation is not a faithful reproduction: it fixes $r = 4$, uses 26 equal-width intervals over the full unit interval, and includes no implementation of the original protocol's complete design or keying details. It is not suitable for securing real data.

Security discussion here is literature review, not a new experiment. Alvarez et al. analyse attacks on an ergodic chaotic cipher [2]. Li et al. discuss problems and countermeasures for Baptista-type systems [3]. We did not reproduce those attacks, countermeasures, or finite-precision error experiments, because the supplied script and results contain only logistic-map simulations and four reversible text checks. The small iteration-count histogram is descriptive and does not test security.

The project therefore demonstrates familiar nonlinear-dynamical behaviour and correct recovery in a toy interval mapping. It does not establish confidentiality, resistance to known- or chosen-plaintext attacks, or equivalence to the cited systems. Modern authenticated, standardized cryptography should be used for real-world information.

References

[1] M. S. Baptista, "Cryptography with chaos," *Physics Letters A*, 240(1-2), 50-54 (1998). doi:10.1016/S0375-9601(98)00086-3

[2] G. Alvarez, F. Montoya, M. Romera, and G. Pastor, "Cryptanalysis of an ergodic chaotic cipher," *Physics Letters A*, 311(2-3), 172-179 (2003). doi:10.1016/S0375-9601(03)00469-9

[3] S. Li, G. Chen, K.-W. Wong, X. Mou, and Y. Cai, "Baptista-type chaotic cryptosystems: Problems and countermeasures," *Physics Letters A*, 332(5-6), 368-375 (2004). doi:10.1016/j.physleta.2004.09.028